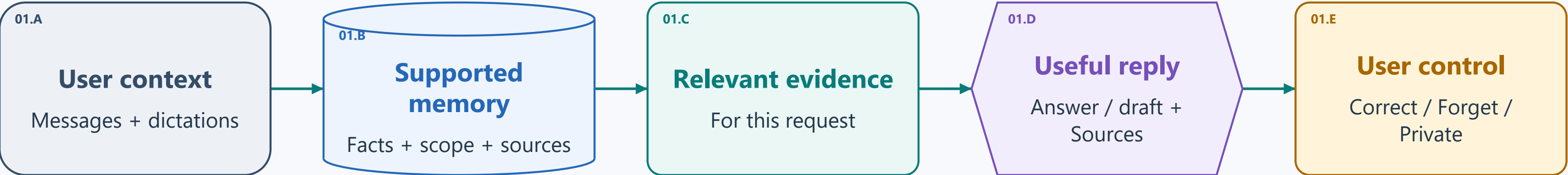


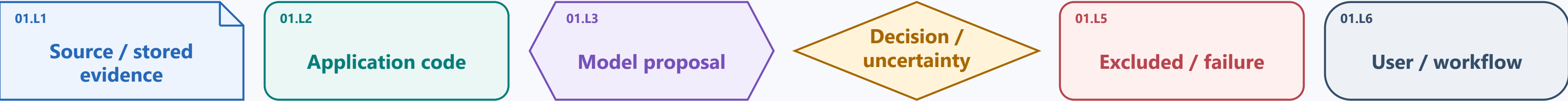
The whole plan, one visual language

Follow the arrows. Use a page + node ID when asking a question.

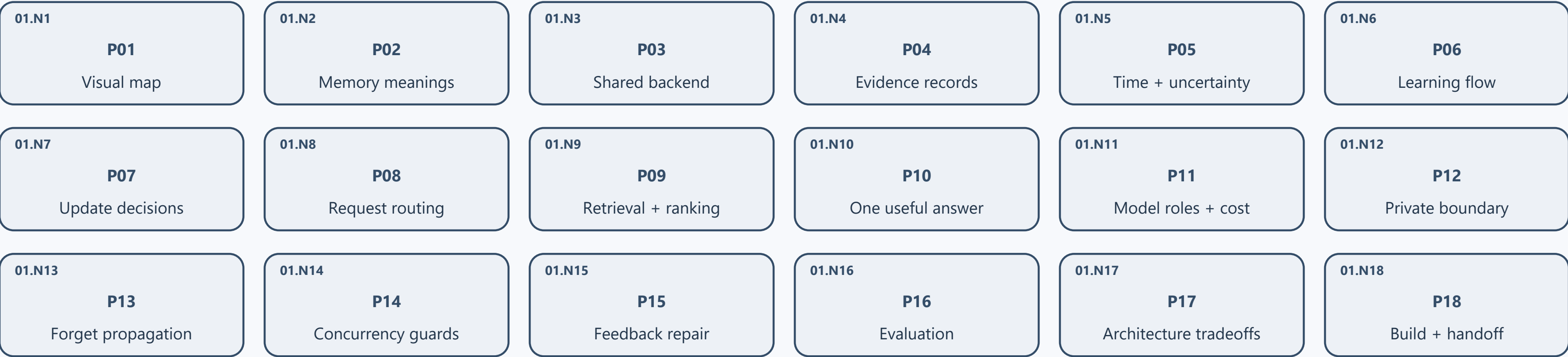
REFERENCE



COLOR = ROLE



CLICK A CARD TO JUMP / SOLID = FLOW / DASHED = OPTIONAL OR CONDITIONAL



STATUS TAGS: DECIDED / PROPOSED / VERIFIED / UNTESTED | Color never means 'already built'.

REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

BRIEF

Assignment scope: pp. 3-5

Includes factual, episodic and preference-level understanding; choose a useful product scope.

PROJECT DECISION

Kivi implementation plan

Proposed milestones, scope cuts, approval boundaries and the minimum complete reviewer journey.

PROJECT DECISION

Kivi requirements and open choices

Defines eligible evidence, Normal/Private, Correct/Forget and proposed implementation choices.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

Semantic memory: assignment scope + terminology

Includes useful episodes; automatic procedural learning is deferred.

PROPOSED

ASSIGNMENT UMBRELLA: "SEMANTIC MEMORY"

02.A

Semantic understanding

Reusable facts + scoped preferences
Mira: three-bullet Atlas updates

INCLUDED

02.B

Episodic understanding

Reported events + their context
User reported sending a checklist

INCLUDED

PLANNED SCOPE: FACTS + PREFERENCES + USEFUL EPISODES

PROCEDURAL / HOW TO ACT

02.C

Automatic procedure learning

Learning reusable task steps
Reviewed code + prompts will guide v1 behavior

DEFERRED

Deferral is our scope choice, not a brief rule.

PERSISTENCE = LIFETIME, INDEPENDENT OF CONTENT KIND

02.D

Persistent store

Survives requests / restart
Can hold different content kinds

retrieve

02.E

Working context

Selected evidence + current request
Fits the model input budget

prompt

02.F

Model response

Uses the supplied context
Does not update model weights

02.G

Categories do not require separate databases

02.H

Saved claim does not mean independently verified

REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

BRIEF

Assignment scope: pp. 3-5

Includes factual, episodic and preference-level understanding; choose a useful product scope.

PRIMARY SOURCE

LangChain memory concepts

Separates facts, experiences and instructions; explains thread state, stores and memory writing.

PROJECT DECISION

Kivi architecture and model proposals

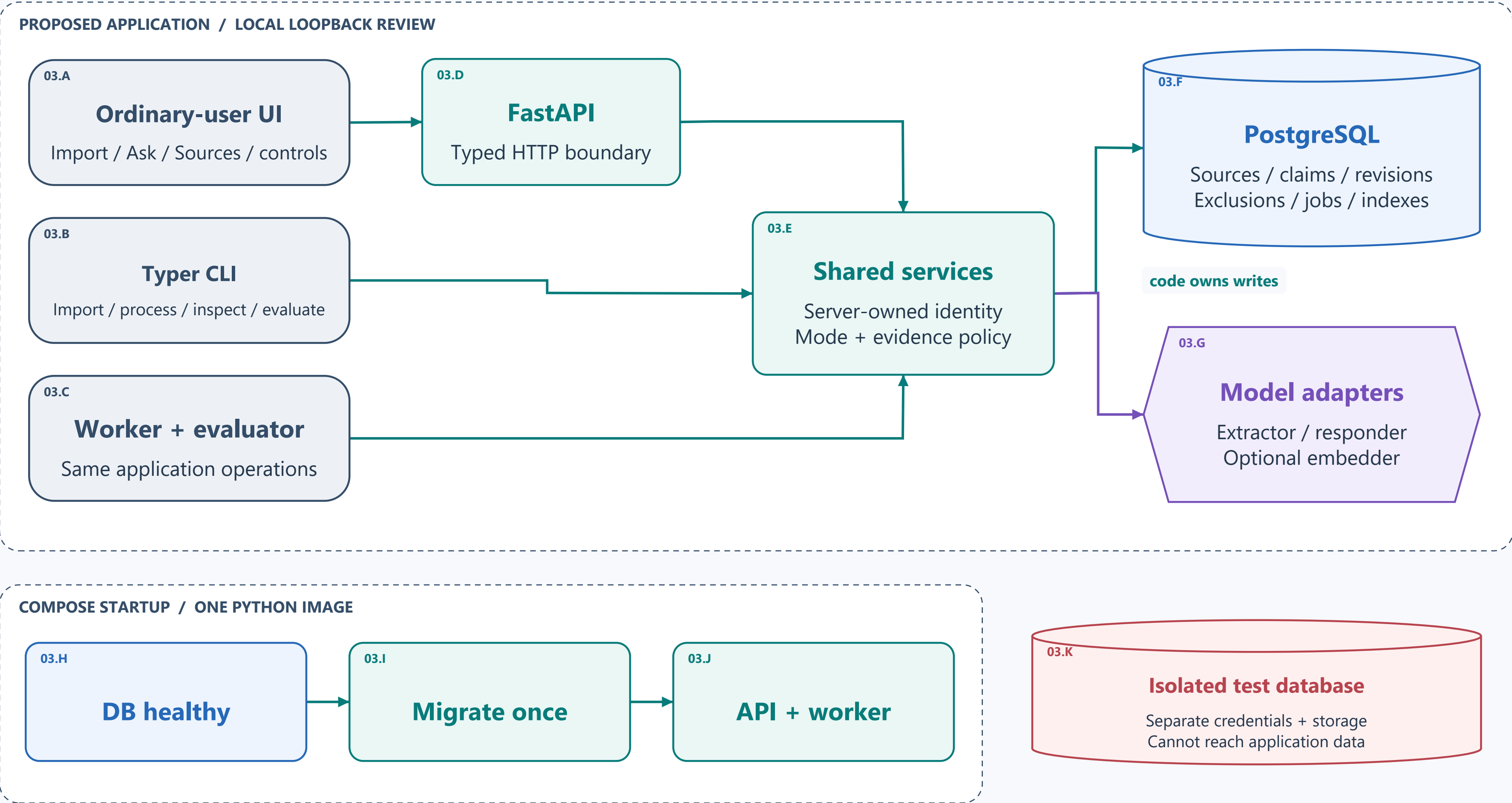
Defines proposed records, pipelines, model boundaries, full shortlist and concurrency guards.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

One backend, several ways to use it

Every entry point follows the same policy and service logic.

PROPOSED



REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PRIMARY SOURCE

Docker Compose startup ordering

Explains dependency ordering, healthy-service conditions and successful one-shot completion.

PRIMARY SOURCE

FastAPI application containers

Explains building and running a FastAPI application in a container image.

PROJECT DECISION

Kivi architecture and model proposals

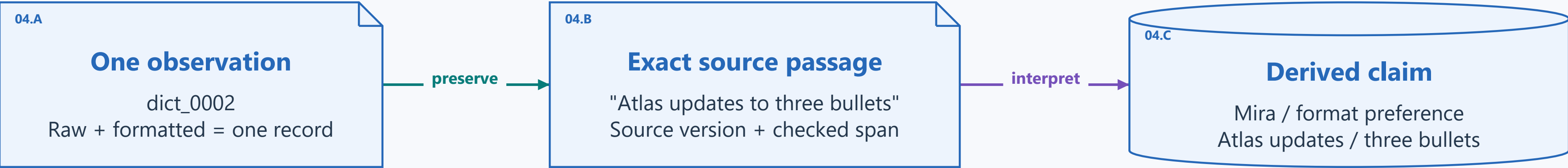
Defines proposed records, pipelines, model boundaries, full shortlist and concurrency guards.

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A memory needs an evidence trail

Store meaning, attribution and uncertainty alongside the original source.

PROPOSED



CLAIM REVISION / INSPECTABLE FIELDS

04.D
Identity + ownership
Whose source? Who said it?
Stable IDs; names are not identity

04.E
Subject + scope
Mira + Atlas updates
Not every recipient or topic

04.F
Typed meaning
Predicate + value + units
Negation + conditions

04.G
Evidence status
Reported / tentative / disputed
Model confidence is not proof

04.H
Time
Capture / import / event / validity
Missing times stay unknown

04.I
Lifecycle + lineage
Active / superseded / corrected
Source -> claim -> index

04.J

Evidence-backed relations
Supersedes / contradicts / conditional-on

04.K

Search index = derived view
Keep original evidence; invalidate stale derivatives

REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PRIMARY SOURCE

W3C PROV: evidence and derivation
Provides concepts for tracing entities, activities, attribution and derived information.

BRIEF

Assignment evaluation: pp. 4-6
Corpus, complete-pipeline evaluation, provenance, supported answers and abstention (pp. 4-6).

PROJECT DECISION

Kivi architecture and model proposals
Defines proposed records, pipelines, model boundaries, full shortlist and concurrency guards.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

Keep time and certainty separate

A newer import can describe an older plan.

PROPOSED

ATLAS / TWO OBSERVATIONS, ONE CHANGING PLAN



Launch date is the claim value. Its effective start is separate; unknown stays unknown.

EVIDENCE STATUS / HOW STRONG IS THE STATEMENT?

05.D

Reported / tentative / disputed

Ravi could own the budget IF Finance signs off
Conditional possibility; no confirmed appointment

LIFECYCLE / HOW MAY THIS VERSION BE USED?

05.E

Active / superseded / corrected / excluded

A tentative claim can still be active
A superseded claim may answer a historical question

05.F

Capture time is not content time

Orion says 25 Sep; capture time + app are unknown

05.G

Latest import is not latest truth

Resolve source meaning + scope + relevant time

REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PRIMARY SOURCE

Bitemporal history

Distinguishes when information applies from when a system records or learns about it.

PRIMARY SOURCE

W3C PROV: evidence and derivation

Provides concepts for tracing entities, activities, attribution and derived information.

PROJECT DECISION

Kivi architecture and model proposals

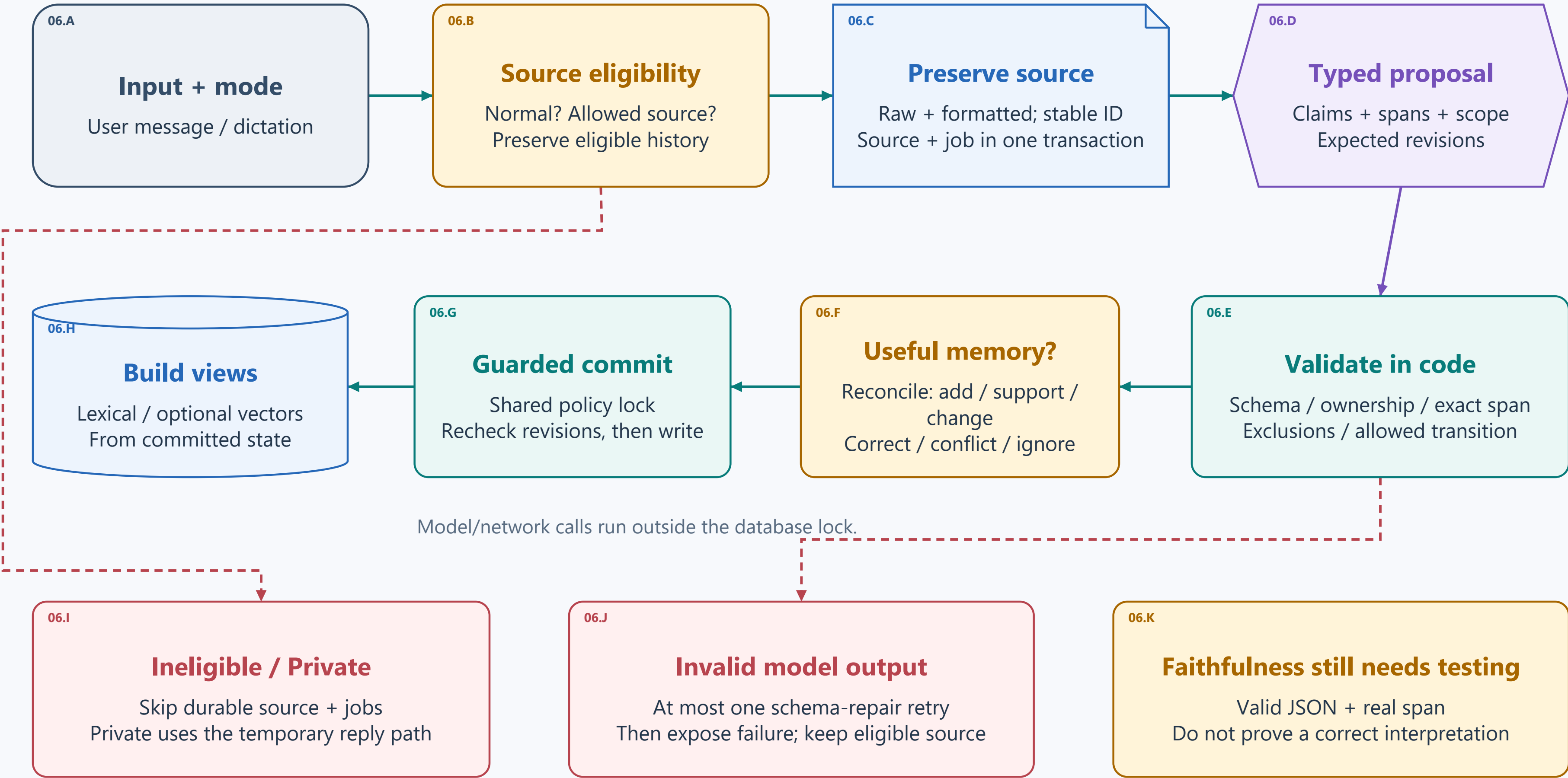
Defines proposed records, pipelines, model boundaries, full shortlist and concurrency guards.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

Learning is a guarded pipeline

A small model interprets eligible input; application code validates and commits.

PROPOSED



REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PRIMARY SOURCE

LangChain memory concepts

Separates facts, experiences and instructions; explains thread state, stores and memory writing.

PRIMARY SOURCE

PostgreSQL explicit locking

Documents transaction locks used to serialize shared policy validation and state changes.

PROJECT DECISION

Kivi architecture and model proposals

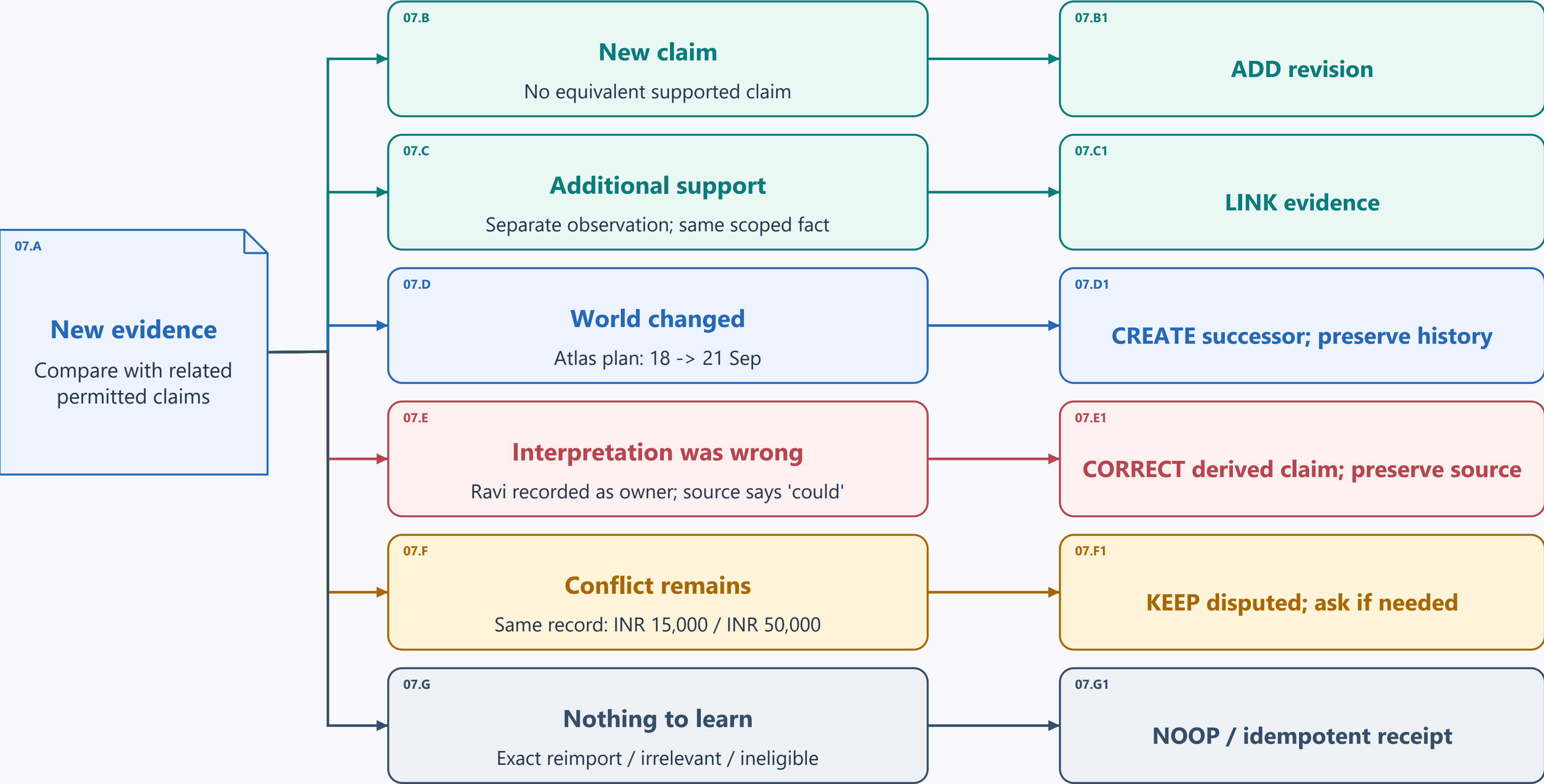
Defines proposed records, pipelines, model boundaries, full shortlist and concurrency guards.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

The same new sentence can mean different things

Compare subject, property, scope, source meaning and time before changing memory.

PROPOSED



Same words can be different observations. Similar names do not establish the same person.

REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PRIMARY SOURCE

Bitemporal history

Distinguishes when information applies from when a system records or learns about it.

PRIMARY SOURCE

W3C PROV: evidence and derivation

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PROJECT DECISION

Kivi architecture and model proposals

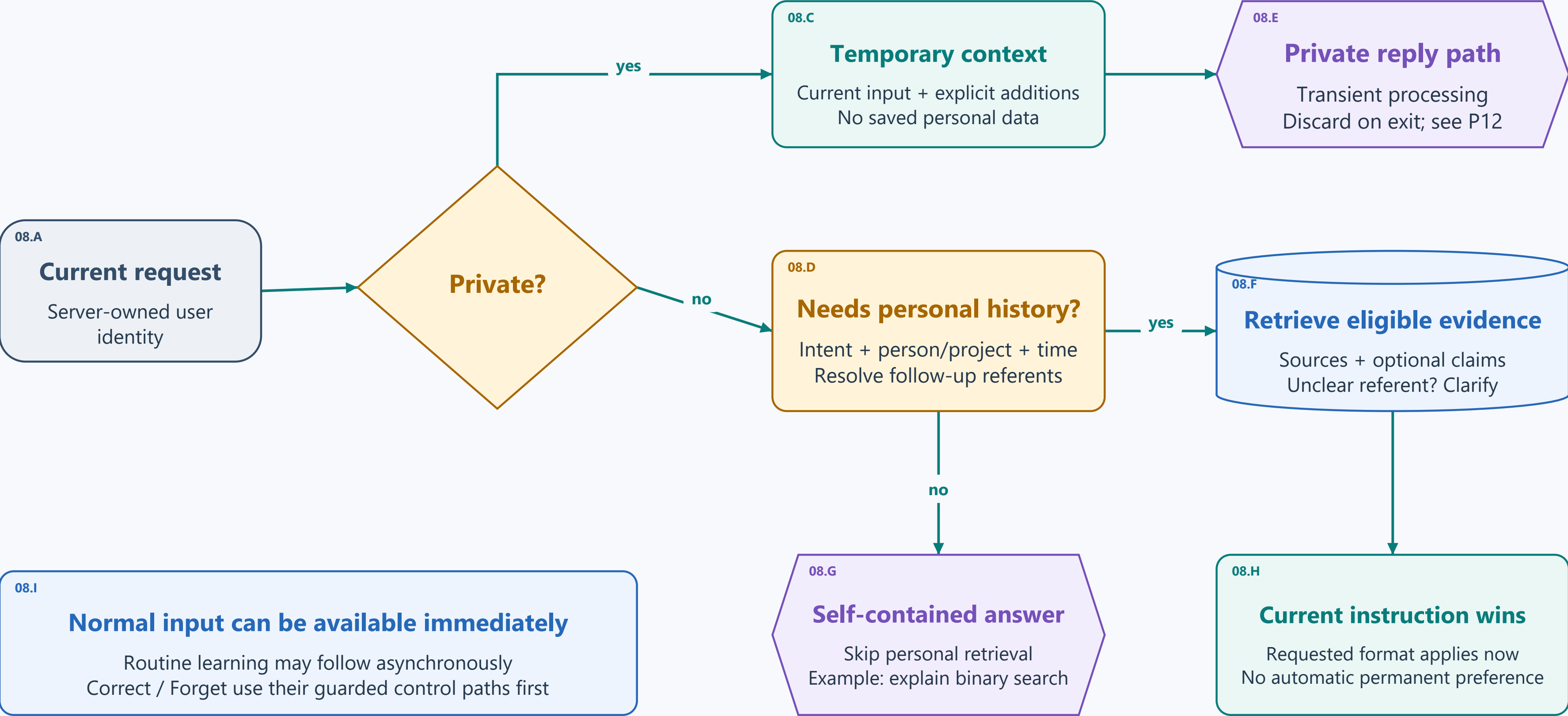
Defines proposed records, pipelines, model boundaries, full shortlist and concurrency guards.

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Decide whether personal retrieval is needed

Check mode before accessing any saved personal information.

PROPOSED



REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PRIMARY SOURCE

LangChain memory concepts

Separates facts, experiences and instructions; explains thread state, stores and memory writing.

PROJECT DECISION

Kivi requirements and open choices

Defines eligible evidence, Normal/Private, Correct/Forget and proposed implementation choices.

PROJECT DECISION

Kivi architecture and model proposals

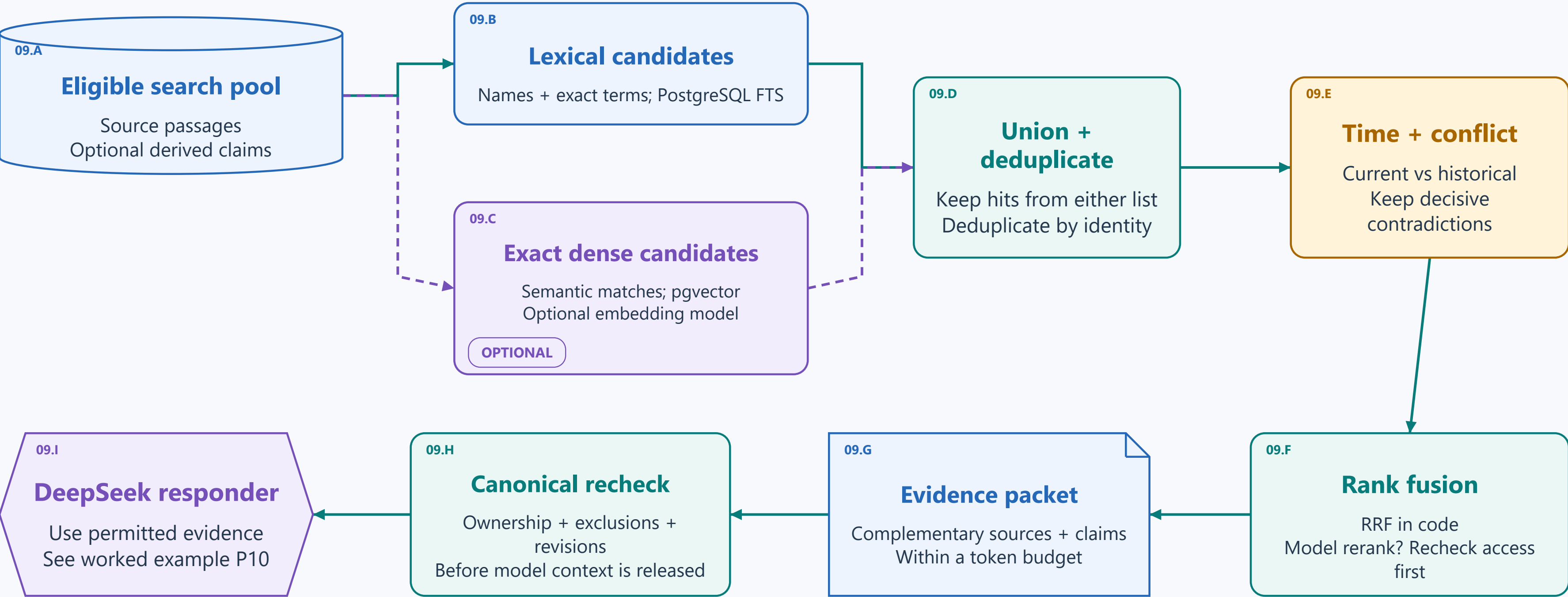
Defines proposed records, pipelines, model boundaries, full shortlist and concurrency guards.

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Retrieve candidates, then rank useful evidence

Similarity finds candidates; provenance and policy determine usable evidence.

PROPOSED



09.J

RRF(d) = $\sum \frac{1}{60 + \text{rank}}$

Only lists containing d; ranks start at 1
Example: ranks 1 + 3 -> 0.03227

09.K

Exact vector scan: $O(N \times d)$

N candidate vectors; d dimensions
Add ANN only after measured need

09.L

Lexical-only hits must enter the pool

Reranking cannot recover unseen candidates
PostgreSQL FTS is not automatically BM25

REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PRIMARY SOURCE

Reciprocal Rank Fusion paper

Combines candidate rankings using reciprocal rank contributions; motivates the k=60 example.

PRIMARY SOURCE

pgvector: exact and hybrid search

Documents exact/approximate vectors and full-text hybrid search with fusion or reranking.

PROJECT DECISION

Kivi architecture and model proposals

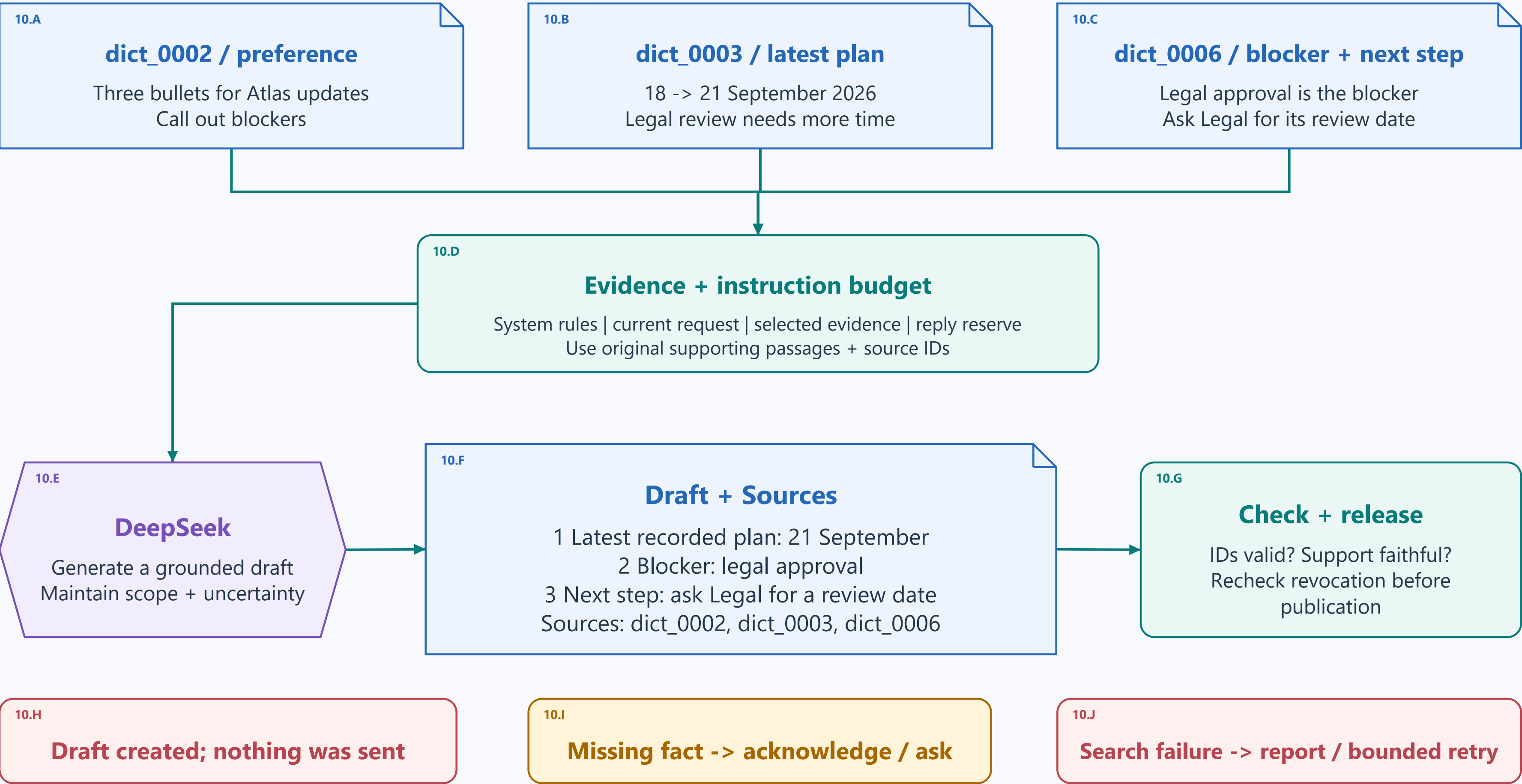
Defines proposed records, pipelines, model boundaries, full shortlist and concurrency guards.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

Watch three memories support one useful draft

Example: 'Draft an Atlas update for Mira in her requested format.'

PROPOSED



REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PRIMARY SOURCE

Anthropic contextual retrieval

Explains preserving chunk context and combining lexical, dense and reranking mechanisms.

PRIMARY SOURCE

W3C PROV: evidence and derivation

Provides concepts for tracing entities, activities, attribution and derived information.

SYNTHETIC EXAMPLE

Synthetic Atlas / Mira fixtures

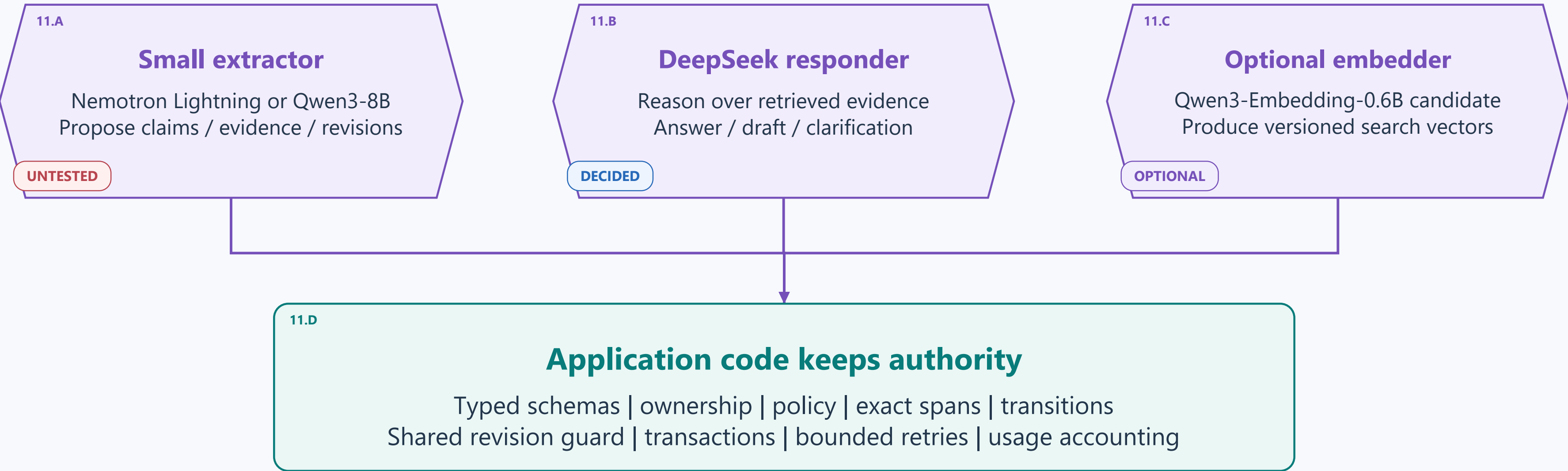
Illustrates changed plans, scoped preferences, reported actions, uncertainty and exact sources.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

Give each model a bounded role

The smaller model is a cost hypothesis to test, not a trusted database operator.

UNTESTED



11.E Before live calls

Verify endpoint + account access
Retention / no-training settings
Credentials + agreed spend ceiling

11.F What a proposal can say

ADD_CLAIM / LINK_EVIDENCE
PROPOSE_SUPERSESSION / NOOP
NEEDS_CLARIFICATION

11.G Measure total cost

Ingestion + answers + retries + failures
Compare cost per successful task
Smaller parameters do not prove savings

REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PRIMARY SOURCE

NVIDIA DeepSeek dated endpoint

Documents the candidate hosted DeepSeek endpoint for the selected main-model role.

PRIMARY SOURCE

Nemotron Lightning model card

Describes a candidate model for bounded typed proposals, including its sparse architecture.

PROJECT DECISION

Kivi architecture and model proposals

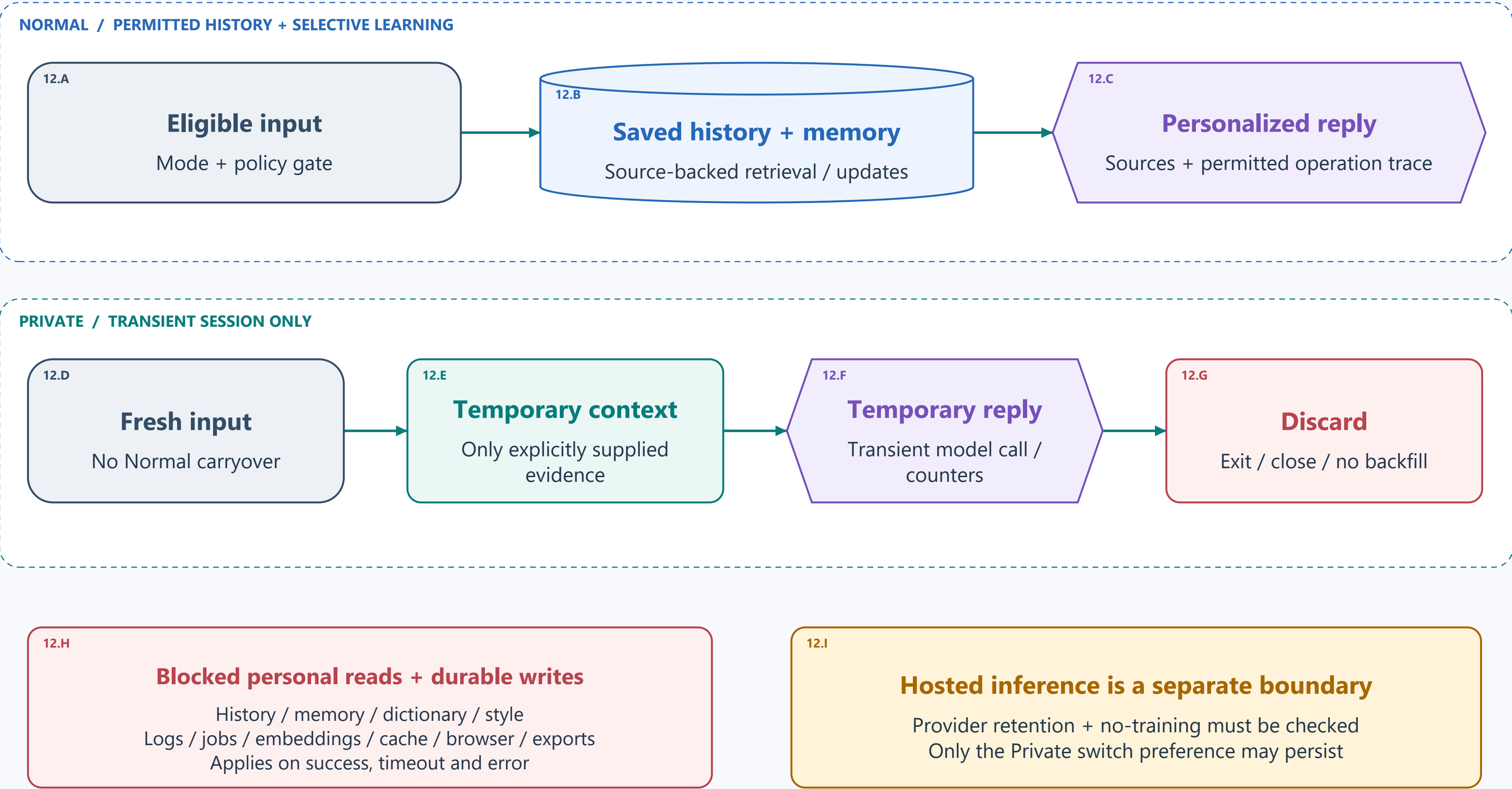
Defines proposed records, pipelines, model boundaries, full shortlist and concurrency guards.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

Private is a boundary before personal state

Private starts fresh and never backfills into Normal.

DECIDED



Private example: saved Atlas date is unavailable unless supplied again as temporary context.

REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PROJECT DECISION

Kivi requirements and open choices

Defines eligible evidence, Normal/Private, Correct/Forget and proposed implementation choices.

PROJECT DECISION

Kivi architecture and model proposals

Defines proposed records, pipelines, model boundaries, full shortlist and concurrency guards.

PROJECT DECISION

Kivi evaluation gates and race tests

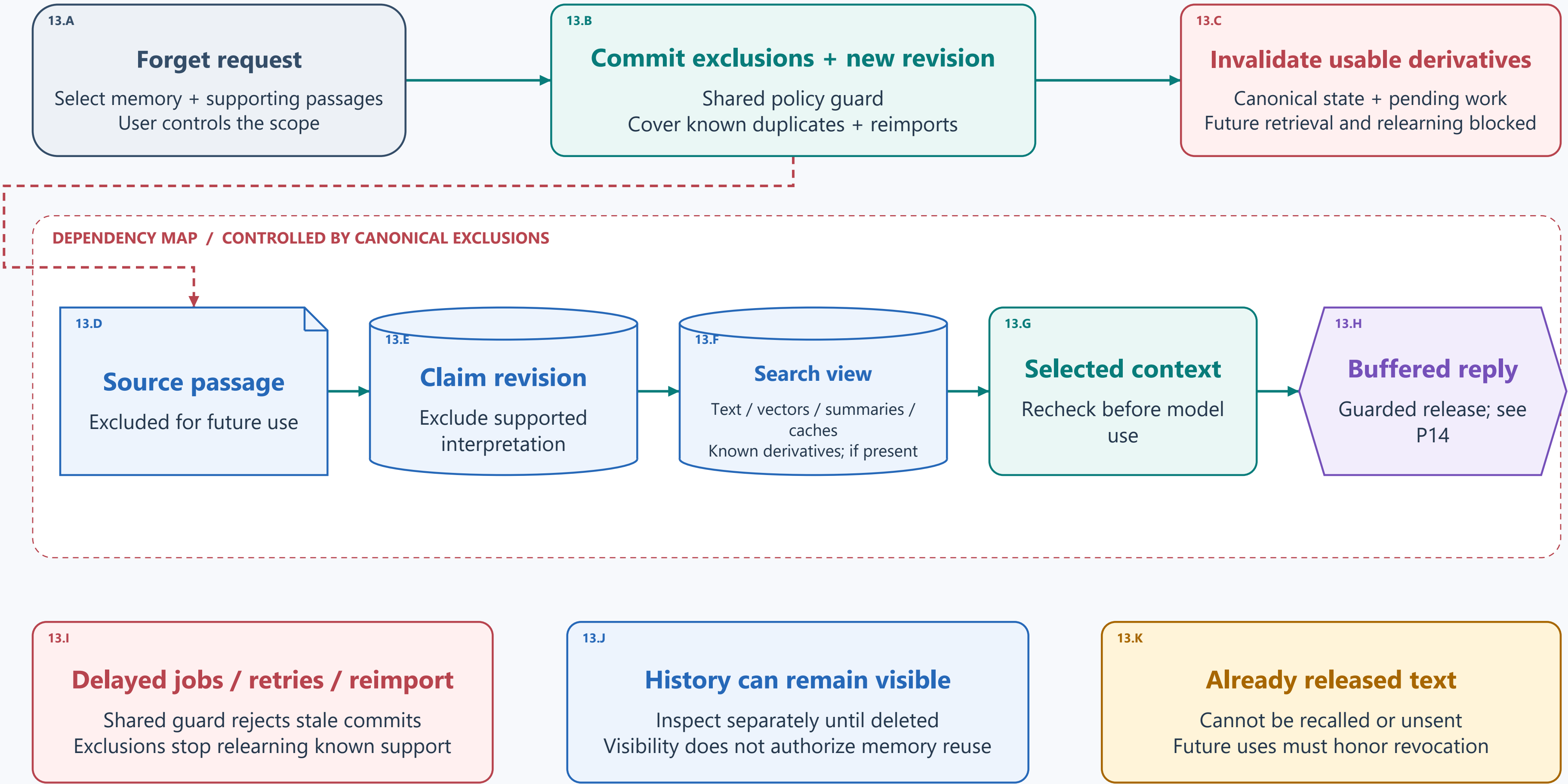
Defines real-DB races, Private checks, repeated live cases, ablations and proposed quality gates.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

Forgetting must follow every dependency

Forgetting a memory is more than deleting its vector.

DECIDED



REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PRIMARY SOURCE

W3C PROV: evidence and derivation

Provides concepts for tracing entities, activities, attribution and derived information.

PRIMARY SOURCE

PostgreSQL explicit locking

Documents transaction locks used to serialize shared policy validation and state changes.

PROJECT DECISION

Kivi requirements and open choices

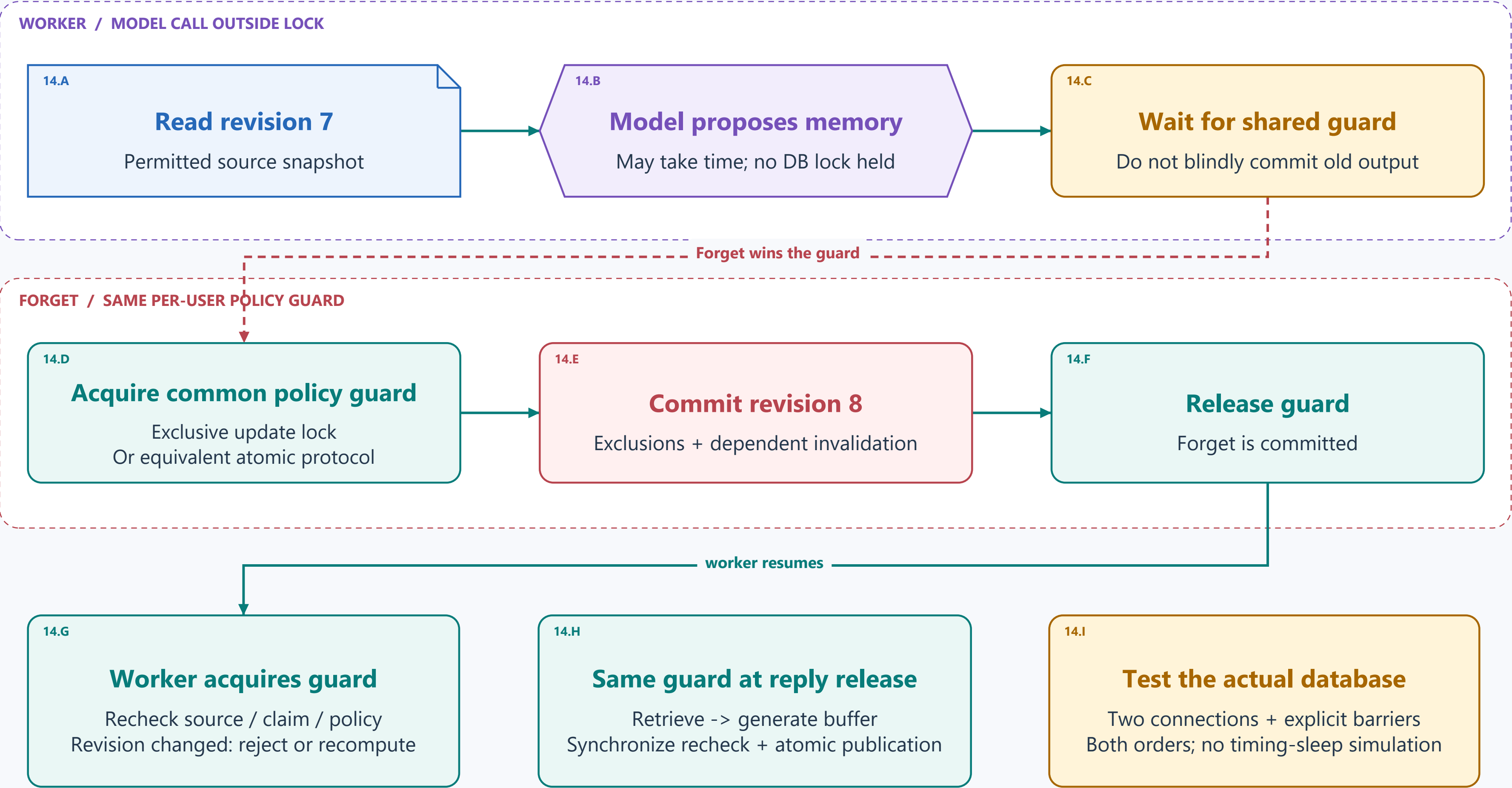
Defines eligible evidence, Normal/Private, Correct/Forget and proposed implementation choices.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

A shared guard defeats stale work

Illustrative interleaving: Forget commits while extraction is in flight.

PROPOSED



REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PRIMARY SOURCE

PostgreSQL explicit locking

Documents transaction locks used to serialize shared policy validation and state changes.

PROJECT DECISION

Kivi architecture and model proposals

Defines proposed records, pipelines, model boundaries, full shortlist and concurrency guards.

PROJECT DECISION

Kivi evaluation gates and race tests

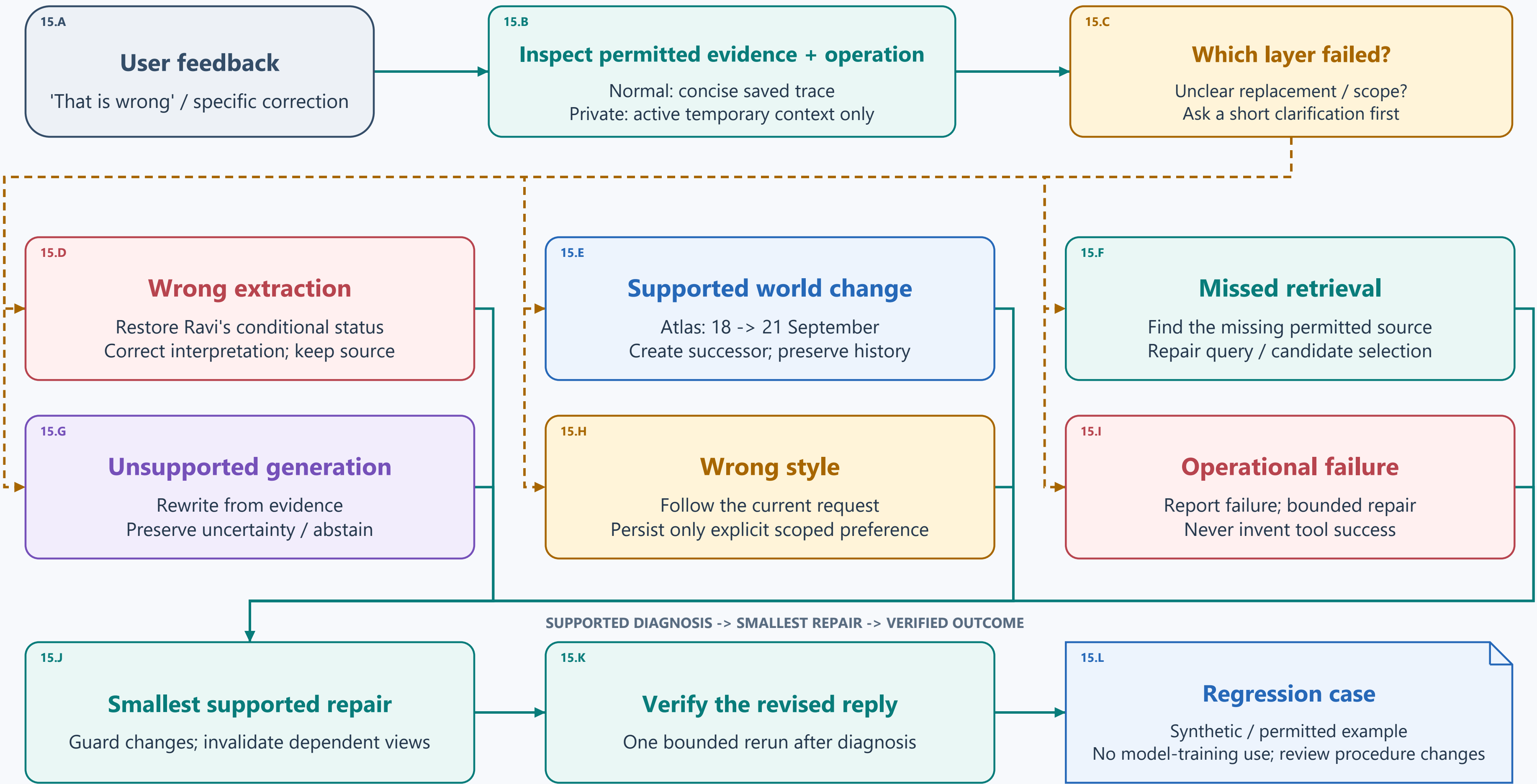
Defines real-DB races, Private checks, repeated live cases, ablations and proposed quality gates.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

Diagnose feedback before changing memory

Dissatisfaction alone does not supply a new fact or a permanent preference.

PROPOSED



REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

PRIMARY SOURCE

Self-Refine: feedback and revision

Studies iterative feedback and refinement of model outputs without additional model training.

PRIMARY SOURCE

Reflexion: verbal feedback memory

Studies agents using linguistic feedback and stored reflections to improve later attempts.

PROJECT DECISION

Kivi architecture and model proposals

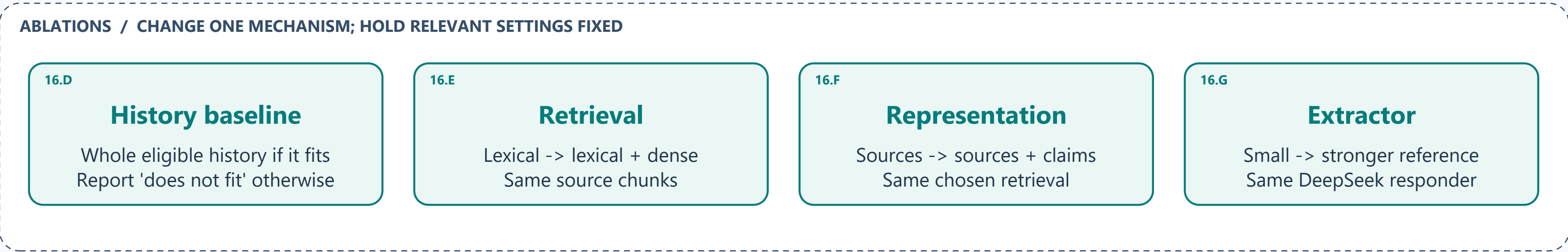
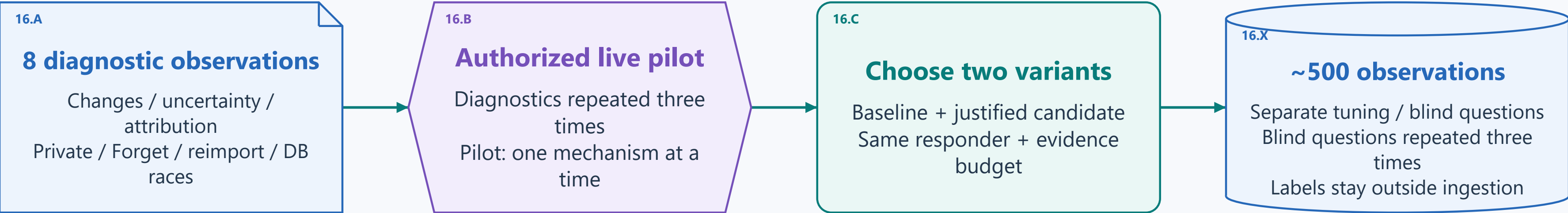
Defines proposed records, pipelines, model boundaries, full shortlist and concurrency guards.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

Prove usefulness with controlled comparisons

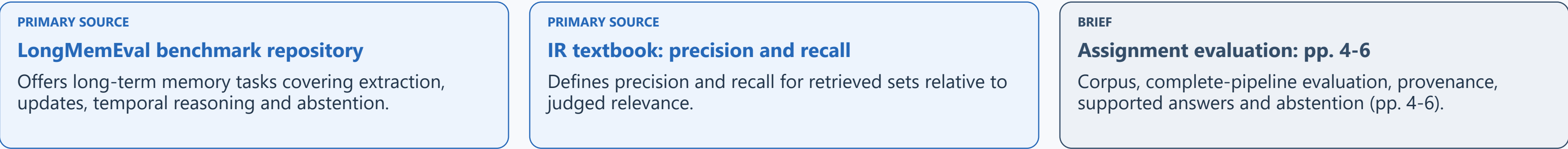
Targets and planned tests are not measured results.

UNTESTED



REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

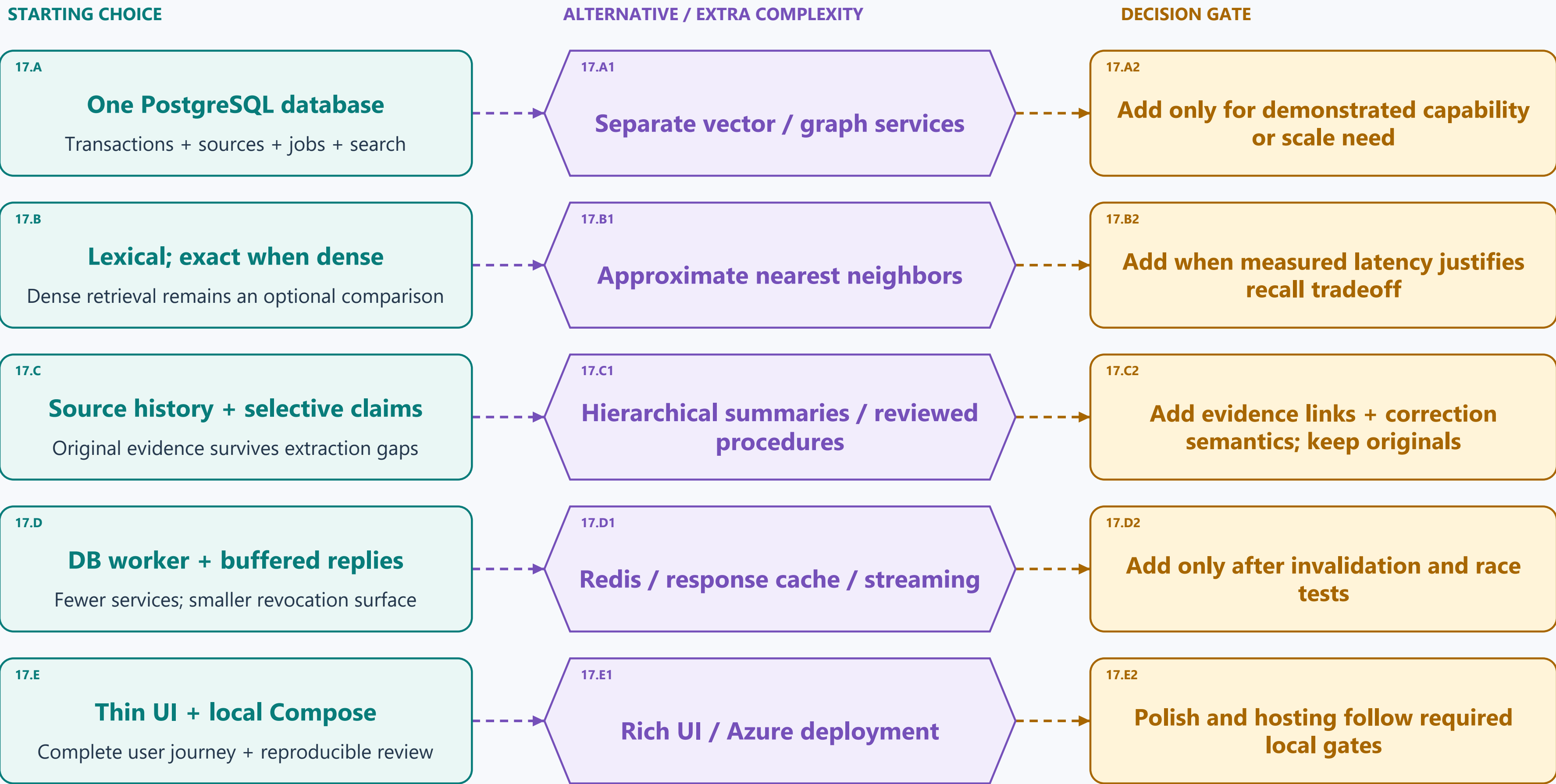


References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

Every extra mechanism must earn its place

Start small. Measure a concrete failure. Add the smallest justified mechanism.

PROPOSED



The goal is a supported useful task, not the largest collection of frameworks.

REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

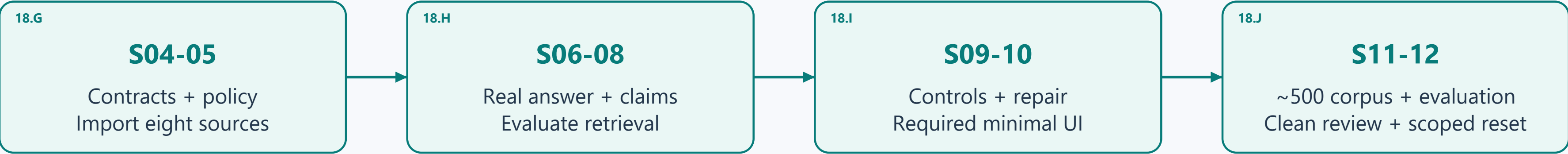
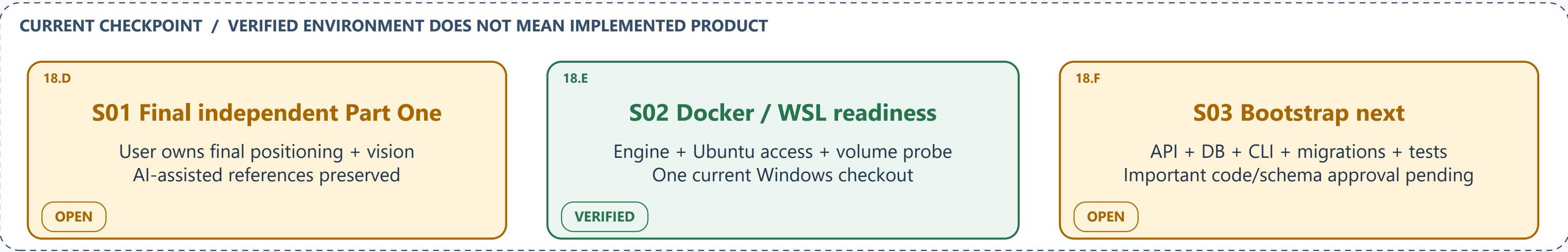
PRIMARY SOURCE pgvector: exact and hybrid search Documents exact/approximate vectors and full-text hybrid search with fusion or reranking.	PRIMARY SOURCE Docker Compose startup ordering Explains dependency ordering, healthy-service conditions and successful one-shot completion.	PROJECT DECISION Kivi requirements and open choices Defines eligible evidence, Normal/Private, Correct/Forget and proposed implementation choices.
---	--	---

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.

Understand here, implement from the same plan

New discussion chat -> recorded decision -> approved CLI work -> tested Git checkpoint.

REFERENCE



REFERENCES / CLICK A CARD

Source register: support, limits + exact brief locators

BRIEF

Assignment product requirements: p. 4
Requires an ordinary-user interface connected to actual state, persistence, retrieval and models.

PROJECT DECISION

Kivi implementation plan
Proposed milestones, scope cuts, approval boundaries and the minimum complete reviewer journey.

PROJECT DECISION

Kivi current status and next action
Tracks completed, ongoing and pending work plus test, commit, push and verification workflow.

References explain methods or record requirements. They do not prove Kivi works. See the register for each source's limits.